

Assessment of atherosclerosis: the role of flow-mediated dilatation.

Evidence suggests that endothelial dysfunction is on the causal pathway for both atherogenesis and destabilization of established plaques. In this review, the role of flow-mediated dilatation (FMD) as a non-invasive method to assess endothelial function is discussed. Technical modifications and development of analysis software have significantly improved the variability of the method. Following a strict standardized protocol enables reproducible measurements to be achieved and export of the technique from specialized laboratories to population studies and multicentre settings. Endothelial function assessed by FMD has been shown to be affected by cardiovascular risk factors, to be related to structural arterial disease and to cardiovascular outcome, validating its use for studying the pathophysiology of arterial disease. Numerous studies have also demonstrated that it is responsive to physiological and pharmacological interventions. Flow-mediated dilatation provides unique opportunities in drug development programmes to assess an early rapidly responsive signal of risk or benefit, complementing endpoints of structural arterial disease and cardiovascular outcomes that take much longer and are more expensive.